

Ans 1.

Inertia is the property of a body due to which it resists any change in its state of rest or of uniform motion in a straight line.

Ans 2.

Newton's third law states that to every action there is an equal and opposite reaction. The action and reaction act on two different bodies.

Ans 4.

Mass is the amount of ~~force~~ matter present in a body and it is measured in kg. Weight is the force with which the earth pulls the body and it is also measured in kg.

Q 3 (b)

Given, $m = 5 \text{ kg}$, $v = 4 \text{ m/s}$

Momentum $p = m \times v$

$$= 5 \times 4 = 20 \text{ kg m/s}$$

3 (a).

The SI unit of momentum is kg m/s .

Ans 6 (ii).

When a bullet is fired from a gun, the gun moves backward. This backward movement is called recoil and it happens because of conservation of momentum.

6 (i).

The law of conservation of momentum says that when no external force acts on a system, the total momentum of the system before collision is equal to the total momentum after the collision.

Ans 5.

The fielder lowers his hands so that the ball takes more time to stop. As the time increases, the rate of change of momentum decreases, so the force on his hands becomes less and he does not get hurt.

Ans 7.

Given : $F = 10 \text{ N}$, $m = 2 \text{ kg}$, $u = 0$, $t = 3 \text{ s}$

We know that $F = m a$

$$\text{So } a = F / m = 10 / 2$$

$$a = 5 \text{ m/s}^2$$

Now using $v = u + at$

$$v = 0 + 5 \times 3 = 15 \text{ m/s}$$

Hence acceleration is 5 m/s^2 and velocity after 3 seconds is 15 m/s .

Ans 8.

When a body moves on a circular path with a constant speed, its motion is called uniform circular motion.

The direction of velocity keeps changing at every point. Example - the motion of the moon around the earth.

Ans 10.

(1) When a bus starts suddenly, the passengers fall backward because the lower part of the body moves with the bus but the upper part remains at rest.

(2) Dust particles come out when we beat a carpet with a stick.

Rough -

$$10 \div 2 = 5$$

$$5 \times 3 = 15$$

Sir, I could not attempt Q9 as the time was over.